

Unit 1, Lesson 3: Isometry



Benchmark

MA.912.GR.2.2 Identify transformations that do or do not preserve distance.



Learning Targets

- ☐ I can define rigid motions.
- ☐ I can show rigid motions by mapping using arrows on the plane.
- ☐ I can explain why isometry preserves distance.



1.3.1 Warm-Up: Animation Flipbook

An animation flipbook is a small book of drawings with a series of pictures that appear to move when the pages are flipped in rapid succession.



The video shows a series of pictures of two clouds. A few screenshots are shown.



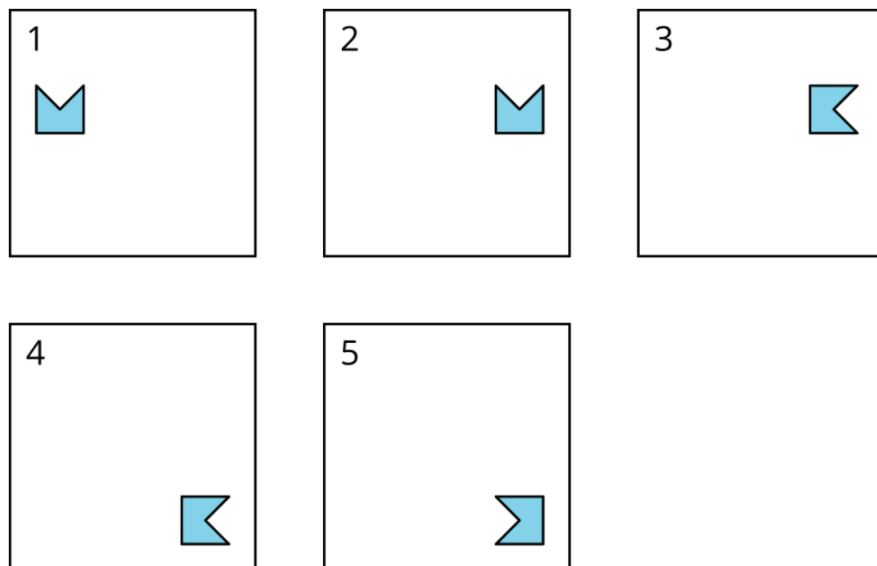
1. What do you notice about the two clouds?
2. At one point the two clouds seemed to move up and down. How do you think the artist was able to do that?



1.3.2 Exploration: Isometry

Work with your partner to complete the following.

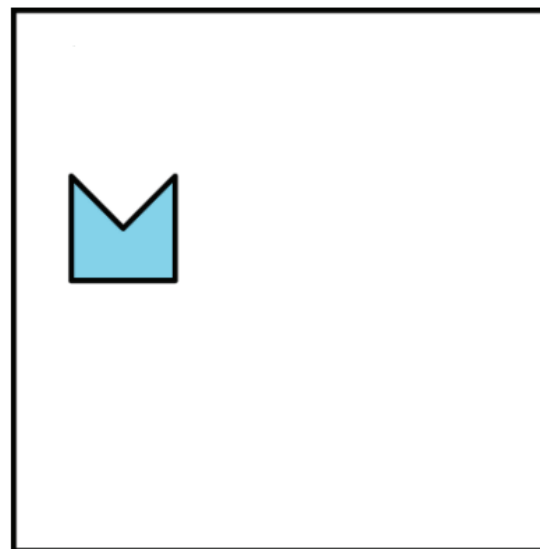
- Five frames that feature a polygon in different positions are shown.



Describe how the polygon moves from one frame to the next.

Movement from...	Description
Frame 1 to Frame 2	
Frame 2 to Frame 3	
Frame 3 to Frame 4	
Frame 4 to Frame 5	

- The diagram shows the position of the polygon from Frame 1.



- Label two pieces of patty paper as **A** and **B**. Then, complete the steps for each piece of patty paper.

Patty paper A

- Copy the polygon.
- Move patty paper **A** to display the polygon as it is shown in Frame 2.

Patty paper B

- Copy the polygon and the square that surrounds it.
- Fold the patty paper vertically so that the top corners of the square match up.
- With the patty paper folded, draw the copy of the polygon onto the other half of the patty paper.

b. Complete the statements.

To draw the polygon as it is shown in Frame 2,

patty paper **A** was moved

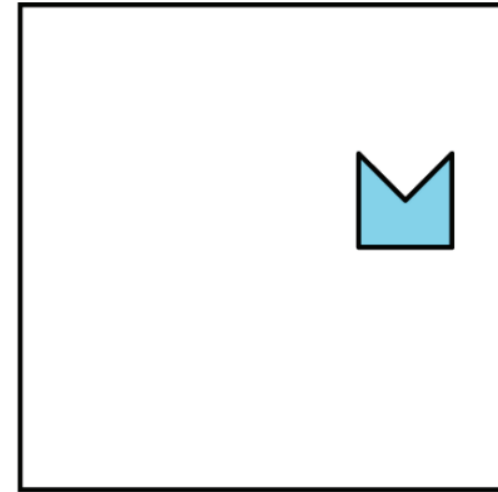
- ☐ up.
- ☐ left.
- ☐ right.
- ☐ down.

When patty paper **B** was folded, the polygon could be traced so that the new polygon is as shown in Frame 2.

The original polygon was ☐ flipped across ☐ rotated around the fold of the paper.

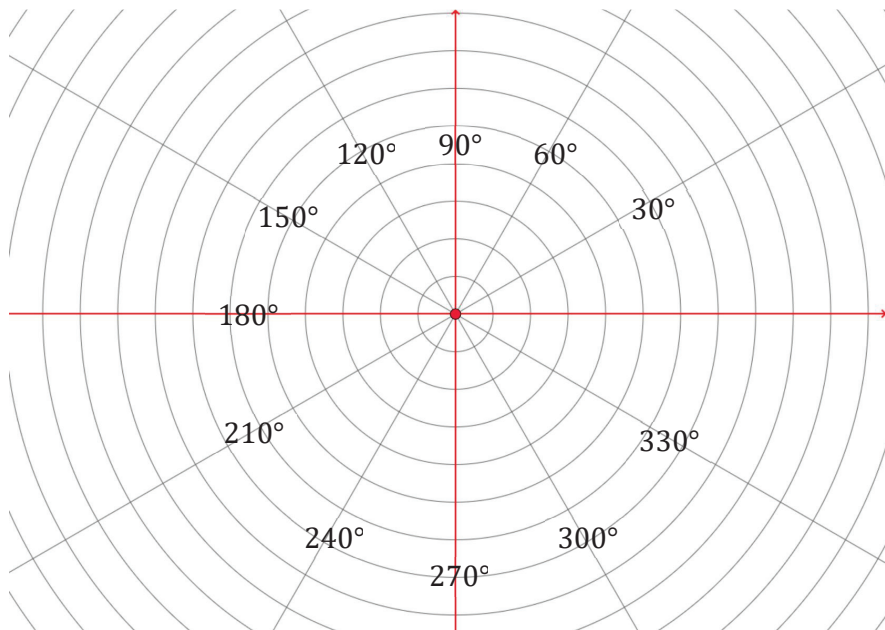
The fold of the paper was the ☐ center of rotation. ☐ line of reflection.

3. The diagram shows where the polygon is in Frame 2.



a. Complete the steps to find the point of rotation.

- Label a piece of patty paper as **C**.
- Copy the polygon onto patty paper **C**.
- Use the point of your writing utensil to find the point of rotation so the polygon can be rotated to look like Frame 3.
- Place a dot on the diagram above to show the point of rotation.
- Place patty paper **C** on the circular graph paper so the point of rotation is where the two red axes meet, and the polygon is oriented as in Frame 2.
- Rotate patty paper **C** so that it looks like Frame 3.



b. Complete the statements.

To move the polygon so it looks like Frame 3, rotate the diagram on patty paper **C** _____ degrees clockwise ↻ or _____ degrees counterclockwise ↺.

The center of rotation is

- the top left vertex.
- the top right vertex.
- the bottom left vertex.
- the bottom right vertex.
- the center of the polygon.

4. Use patty paper, a ruler, or a protractor as necessary to complete the following.

a. Complete the table by determining if the polygon changed in size or shape as it moved.

Movement from...	Did the size change?	Did the shape change?
Frame 1 to Frame 2		
Frame 2 to Frame 3		
Frame 3 to Frame 4		
Frame 4 to Frame 5		

b. What tools or information were used when determining if the size changed?

c. What tools or information were used when determining if the shape changed?

d. Discuss with your partner if the measurement of angles in the polygon changed from Frame 1 to Frame 5.



1.3.3 Bring It Together: Definitions

The movements of the polygon in the five frames are known as reflection, rotation, and translation. During the lesson a reflection was referred to as a flip, which is an informal definition. Another informal definition used was “is moved,” which is formally known as a translation.



A **reflection** is a transformation that produces the mirror image of a geometric figure across a line of reflection.



A **rotation** is a transformation of a figure by turning it about a center point or axis. The amount of rotation can be expressed in the number of degrees. The direction of the rotation for two-dimensional figures can be expressed as clockwise or counterclockwise.



A **translation** is a transformation in which every point in a figure is moved in the same direction and by the same distance.

Isometry is a transformation that preserves distance and angle measure.

1. Discuss with your partner the definition of isometry. Determine which transformations — reflection, rotation, or translation — preserve distance and angle measure. Summarize your discussion.



A **rigid transformation** is a transformation of points in space consisting of a sequence of one or more translations, reflections, or rotations. Rigid transformations preserve distances and angle measures (congruency).



A rigid transformation is also called a **rigid motion**. Rigid means that something is not able to be changed. Isometry is composed of rigid motions.

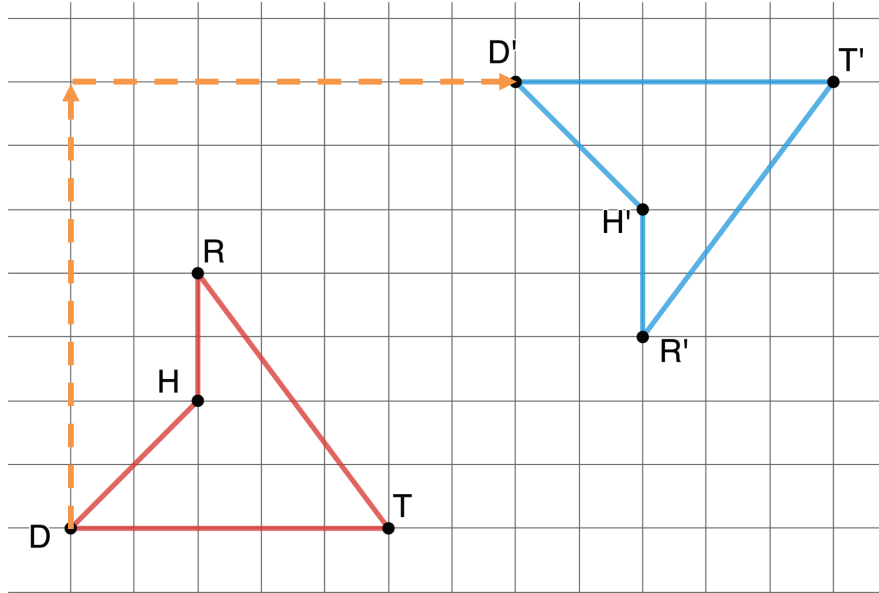


Figures having exactly the same shape and size are **congruent**. Equivalently, two figures are congruent if one can be mapped to the other using a rigid transformation.



1.3.4 Collaboration: Isometry

1. A diagram of $RTDH$ and $R'T'D'H'$ is shown. Arrows are drawn from vertex D to vertex D' .



$R'T'D'H'$ is read as "r prime, t prime, d prime, h prime". This notation is often used when a figure has been transformed.

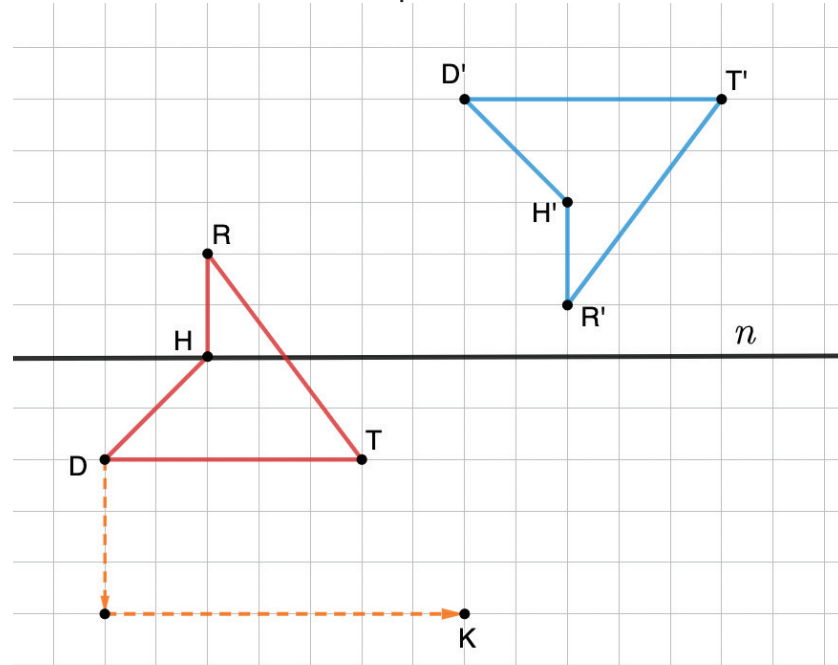
The arrows show a possible path of movement. For the movement to be correct, the same path should exist for all points on a diagram.

Discuss with your partner if all of the points on $RTDH$ can be moved using the same path as point D .



A **mapping** is a standardized form of showing a transformation.

2. A diagram of $RTDH$ and $R'T'D'H'$ is shown. Arrows are drawn from vertex D to point K .



- Draw a line segment from point K to line n . What is the length of the line segment?
- Draw a line segment from point D' to line n . What is the length of the line segment?
- Describe the relationship each line segment has to line n .



If corresponding points have the same distance from a line, then the line is the “**line of reflection.**”

- d. For each of the other vertices on $RTDH$, draw a new point using the same movements done to map point D to point K . Then show the reflection to $R'T'D'H'$ using line segments.
- e. Discuss with your partner if the mapping of **all** of the points on $RTDH$ can be done using the same path and reflection, or if the mapping only works for the vertices. Summarize your discussion.



1.3.5 Wrap-Up: Reflection

1. Complete the table.

The terms and information used in the exploration that I knew were ...	Something I wondered during the exploration was ...	Something I learned during the exploration was ...